

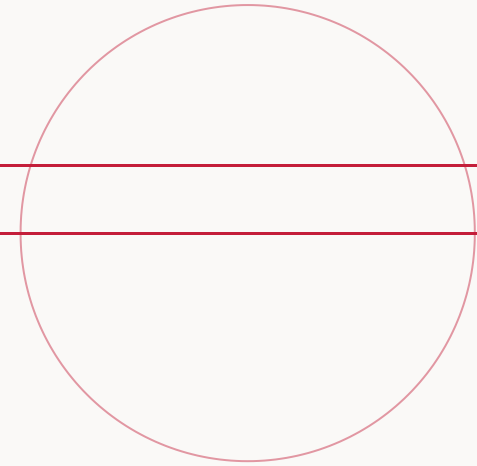
Fama-French 3-Factor Model

*Beyond the market —
size, value, and what's left for skill.*

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A Course on Quantitative Finance



MOTIVATION

CAPM was a start, not the answer

CAPM said one number — β — explains the systematic part of a stock's *return* and *risk*. Decades of evidence say otherwise. Small-caps and value stocks earn premia the market alone cannot account for.

- Add a **size** factor — do small firms beat big firms?
- Add a **value** factor — do cheap firms beat glamour firms?
- What does the stock's α look like once those tilts are stripped out?
- How much more of the variance can three factors explain than one?

One stock, monthly, against the Indian factor file

— 3 —

Reliance Industries, monthly returns aligned with the IIMB / NSE Fama-French factor series. Everything is already in *percent* so units agree across regressors.

STOCK

Reliance

$n = 55$

Monthly excess returns

WINDOW

2021–2025

$\sim 4.5y$

Jun 2021 to Dec 2025

FACTORS

MF · SMB · HML

$+ RF$

Indian FF series, in %

One file, one stock — change a single line in the notebook to re-run for any other ticker.

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CHAPTER I

The three factors — market, size, value

FACTOR 1 · THE MARKET

MKT — the market excess return

The same workhorse from CAPM — the broad-market return, net of the risk-free rate. In the Indian FF file it is the column *MF*.

MARKET FACTOR

$$\text{MKT}_t = r_{m,t} - r_{f,t}$$

Compensation for bearing undiversifiable market risk — the part you can't escape by holding more stocks.

*Already in excess form — do **not** subtract RF a second time.*

FACTOR 2 · SIZE

SMB — Small Minus Big

A long-short portfolio: **buy** small-cap stocks, **sell** large-cap stocks. The return is the *size premium* — what small firms earn over large firms in a given month.

SIZE FACTOR

$$\text{SMB}_t = r_{\text{small},t} - r_{\text{big},t}$$

$\beta_{\text{SMB}} > 0$: the stock behaves like a small-cap. $\beta_{\text{SMB}} < 0$: it behaves like a large-cap.

A factor is just a **portfolio return** — not a magic variable.

FACTOR 3 · VALUE

HML — High Minus Low

Another long-short portfolio, sorted by book-to-market: **buy** high-B/M (value) stocks, **sell** low-B/M (growth) stocks. The return is the *value premium*.

VALUE FACTOR

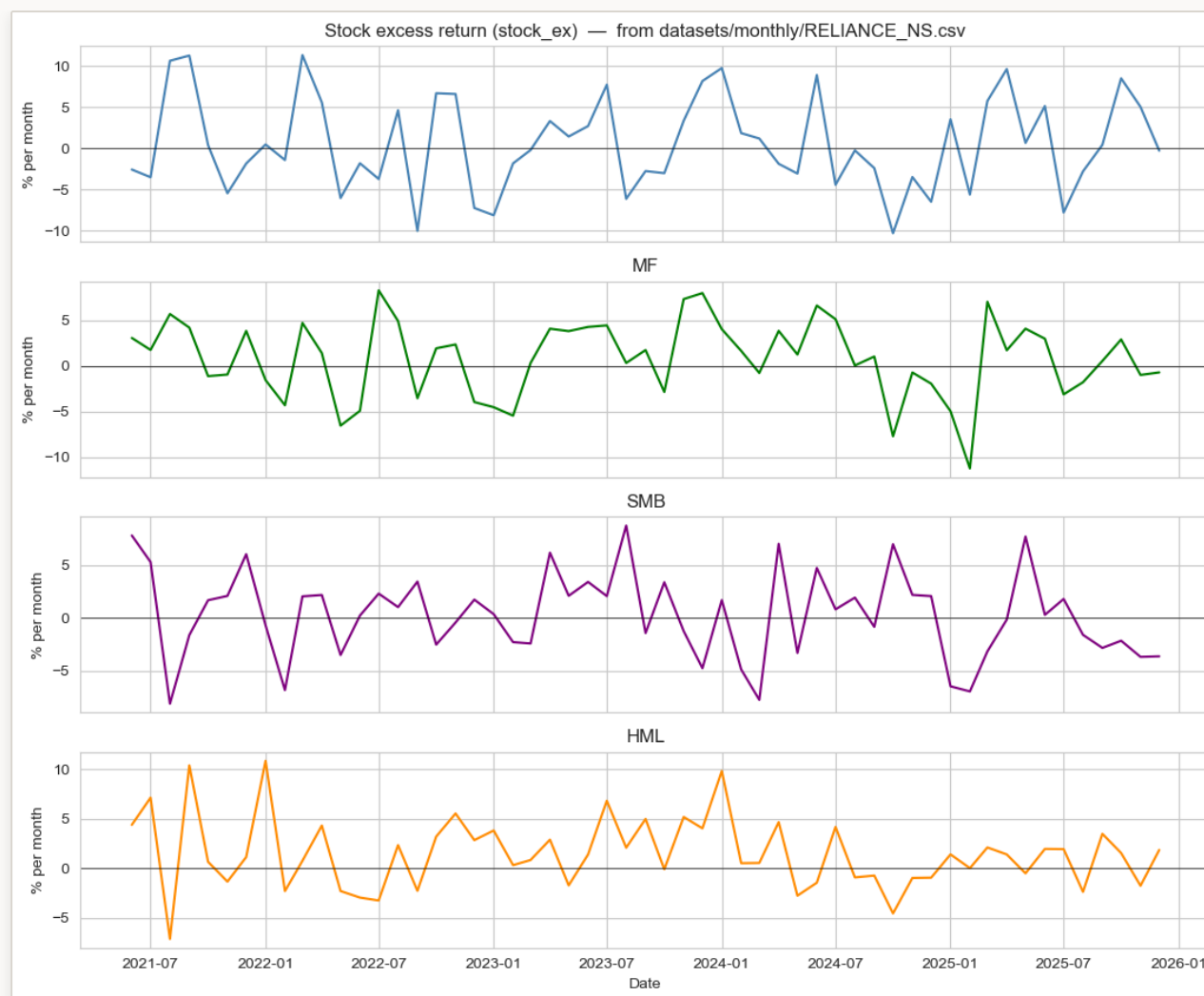
$$\text{HML}_t = r_{\text{value}, t} - r_{\text{growth}, t}$$

$\beta_{\text{HML}} > 0$: tilted toward value. $\beta_{\text{HML}} < 0$: tilted toward growth / glamour.

Three factors — one for the tide, two for the tilts.

Reliance excess return & the three factors

- ▶ Monthly series, all in **percent**
- ▶ MF carries the **biggest swings**
- ▶ SMB & HML wander around **zero**

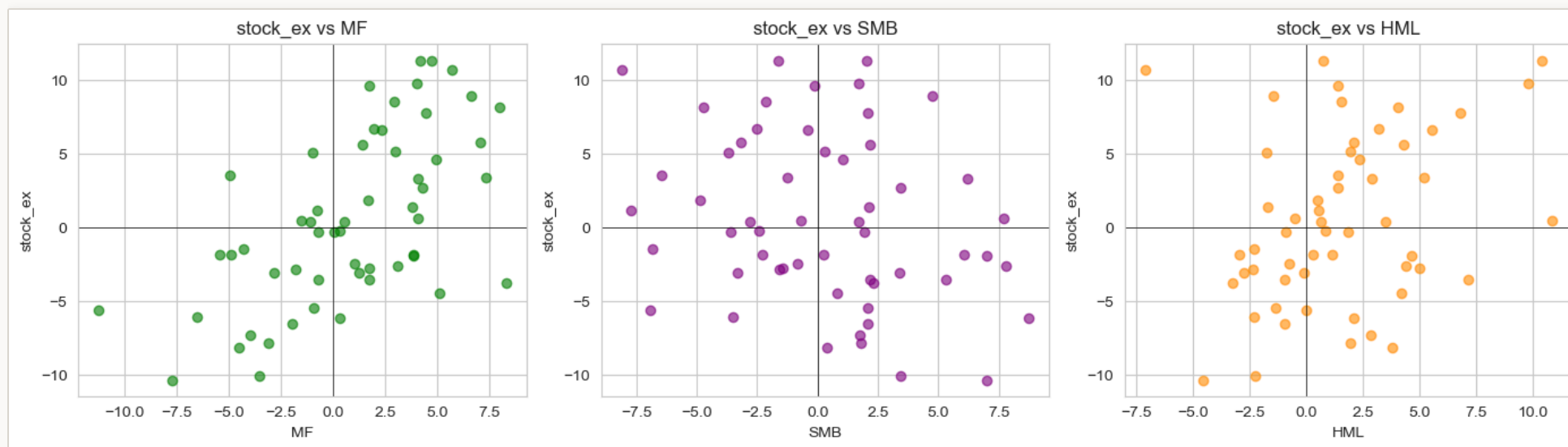


Stock excess return vs each factor

▸ vs MF: **upward** cloud ($\rho \approx +0.61$)

▸ vs SMB: **downward** cloud ($\rho \approx -0.26$)

▸ vs HML: **mild upward** cloud ($\rho \approx +0.29$)



CHAPTER II

The regression
— *CAPM, then FF₃*

BASELINE

CAPM — the one-factor regression

Start from where Module 4 left off — regress the stock's excess return on the market factor alone. The intercept is α ; the slope is β_{MKT} .

MARKET MODEL

$$r_{i,t} - r_{f,t} = \alpha + \beta_{\text{MKT}} \text{MKT}_t + \varepsilon_t$$

For Reliance: $\hat{\alpha} = -0.13\%/mo$ ($p = 0.84$), $\hat{\beta}_{\text{MKT}} = 0.84$ ($p < 0.001$), $R^2 = 0.375$.

The market explains ~38% of Reliance's variance. Two-thirds of the wiggles are still unexplained.

THE MODEL

FF3 — three slopes, one intercept

Same OLS regression, three regressors instead of one. Each β reads as a **partial** exposure — the move in the stock per 1% in that factor, *holding the other two constant*.

FAMA-FRENCH 3-FACTOR REGRESSION

$$r_{i,t} - r_{f,t} = \alpha + \beta_{\text{MKT}} \text{MKT}_t + \beta_{\text{SMB}} \text{SMB}_t + \beta_{\text{HML}} \text{HML}_t + \varepsilon_t$$

α is now the leftover after MKT, size, and value tilts have all been priced — a much stricter bar.

n = 55, k = 3 regressors plus intercept $\Rightarrow n - k - 1 = 51$ degrees of freedom.

RESULTS · OLS VIA STATSMODELS

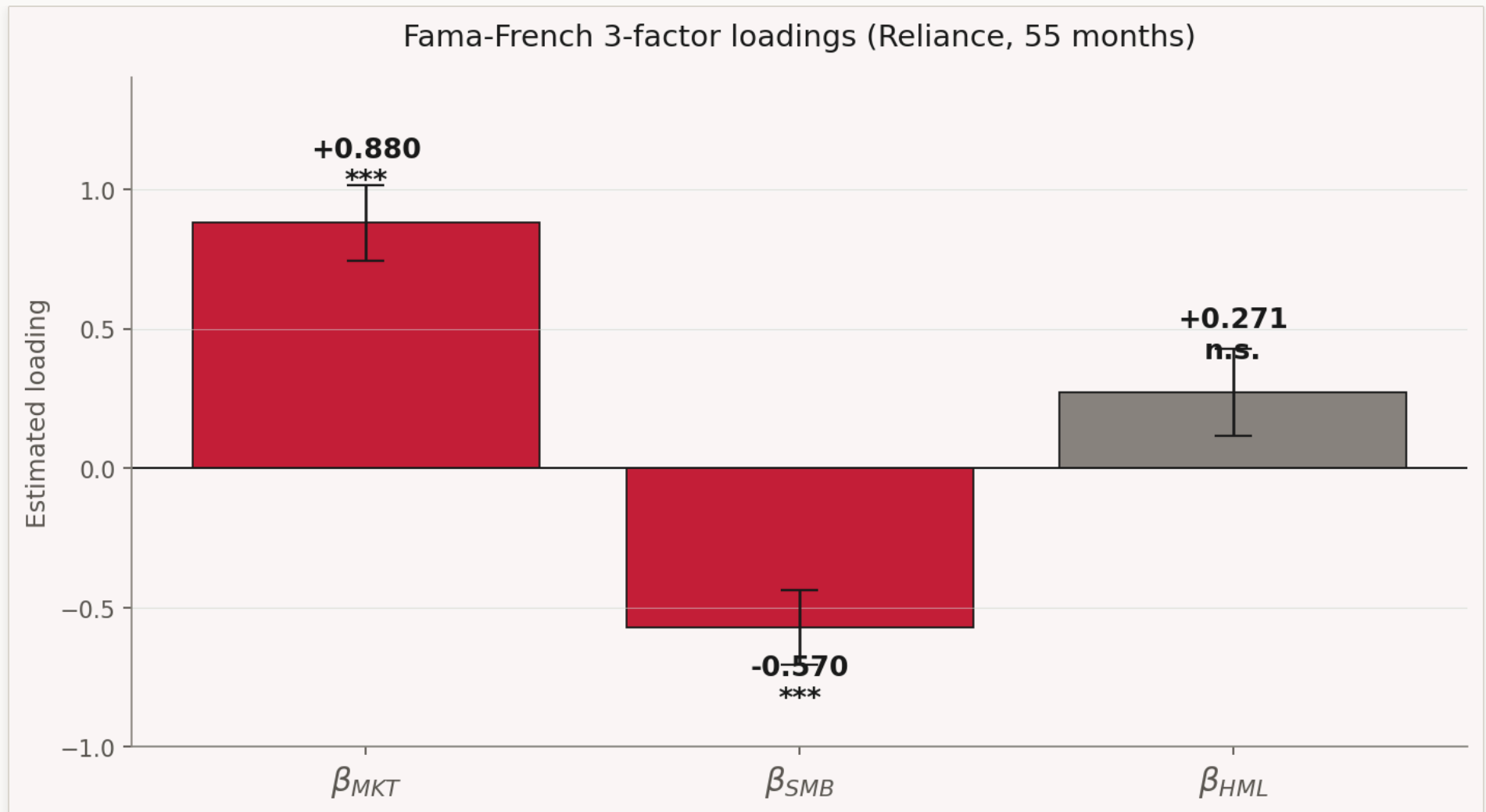
CAPM vs FF₃ — head to head

	A (MONTHLY)	B _{MKT}	B _{SMB}	B _{HML}	R ²	ADJ R ²
CAPM	−0.13%	0.837	—	—	0.375	0.363
FF ₃	−0.39%	0.880	−0.570	+0.271	0.551	0.525

*Adding two factors lifts **R² from 0.375 to 0.551** and **Adj R² from 0.363 to 0.525** — the size and value tilts are doing real work, not just inflating R² by being there.*

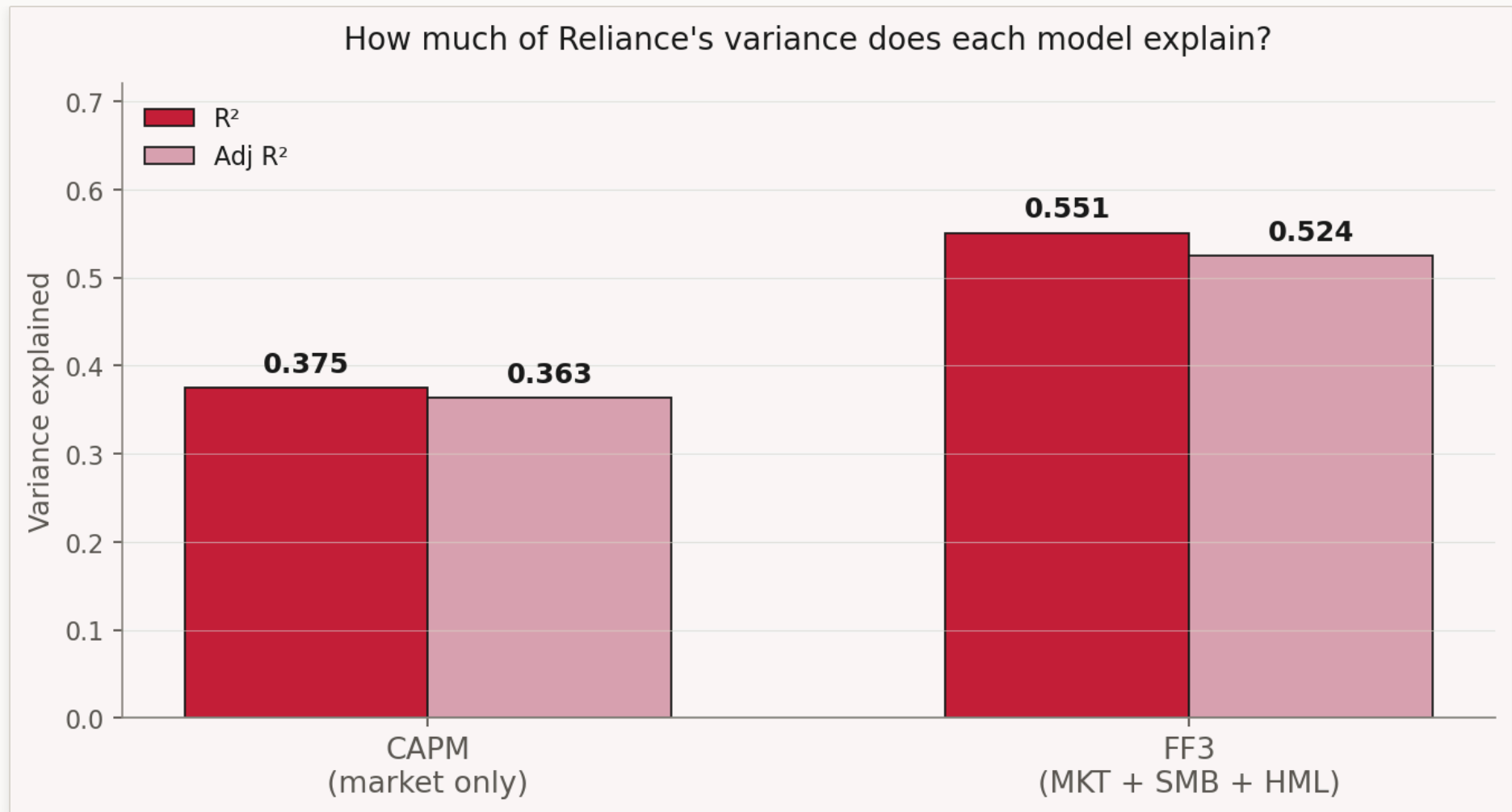
FF₃ loadings for Reliance

▸ $\beta_{\text{MKT}} = +0.88$ — mildly defensive ▸ $\beta_{\text{SMB}} = -0.57$ — large-cap tilt ▸ $\beta_{\text{HML}} = +0.27$ — faint value tilt (n.s.)



Variance explained — CAPM vs FF₃

- ▶ R^2 jumps +17.6 pp with two factors
- ▶ Adj R^2 jumps +16.1 pp — not free
- ▶ Half the wiggles are still idiosyncratic



Reading the four numbers

MARKET

 β_{MKT} **+0.88***Defensive — below 1.*

SIZE

 β_{SMB} **-0.57***Large-cap behaviour.*

VALUE

 β_{HML} **+0.27***Faint value tilt ($p = 0.09$).*

ALPHA

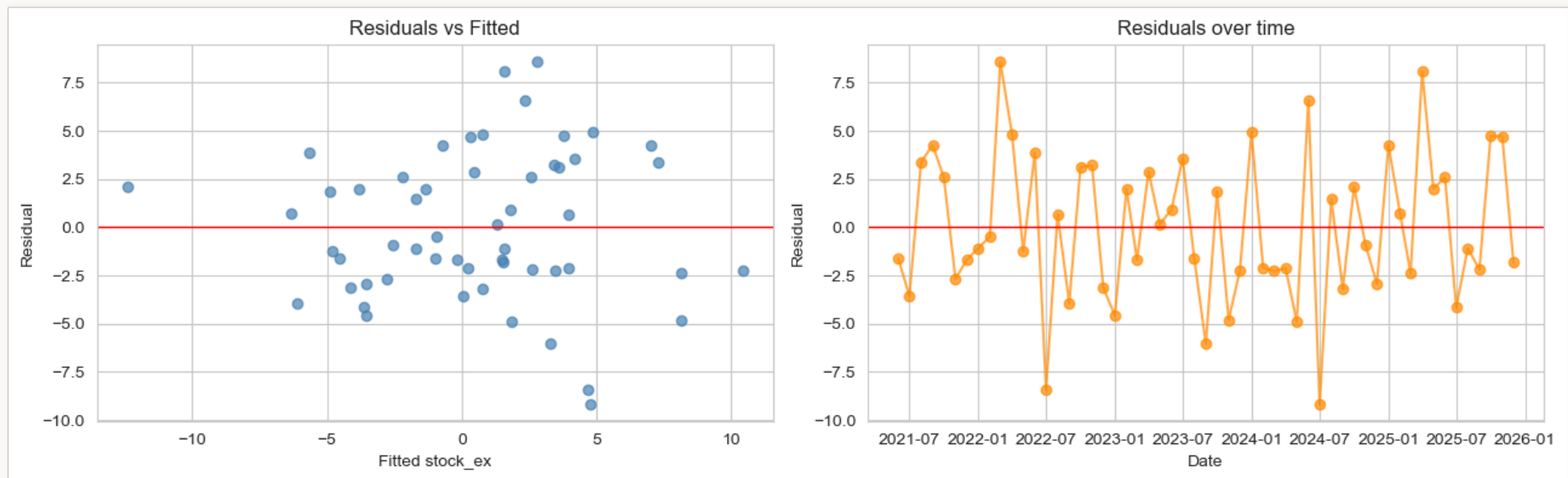
 $\hat{\alpha}$ **-0.39%***Per month, $p = 0.50$.*

Reliance is a **defensive large-cap with a faint value tilt** — once those exposures are priced, no α remains.

DIAGNOSTICS

Residuals vs fitted & over time

- ▶ Cloud is **centred on zero**
- ▶ No obvious **trumpet** or curvature
- ▶ No clustered **regime** over time



HYPOTHESIS TEST

Alpha is hard — 55 months is rarely enough

Even after stripping market, size and value tilts, the question is whether the leftover α is *distinguishable from zero*.

T-STATISTIC FOR α

$$t = \frac{\hat{\alpha}}{\text{SE}(\hat{\alpha})} \sim t_{n-4} \quad \text{under } H_0: \alpha = 0$$

Four parameters estimated $\Rightarrow n - 4 = 51$ degrees of freedom for the FF3 fit.

Reliance: $\hat{\alpha} = -0.39\%$, $SE = 0.58\%$, $t = -0.68$, $p = 0.50$ — far from rejecting $\alpha = 0$. The stock is fairly priced for what it is.

CLOSING

Five things to remember

- **FF3 = CAPM + size + value** — two extra long-short portfolios on the right-hand side.
- Each β is now a *partial* exposure — per 1% in that factor, holding the others fixed.
- R^2 almost always rises; **Adj R^2** is the honest comparison.
- FF3 α is a much stricter bar than CAPM α — market, size and value have all been priced first.
- With ~55 monthly observations, **α is rarely significant** — the data isn't long enough to claim skill.

Three factors price the tilts.

— *what's left is either skill, or noise. Usually noise.*